

Software Requirements Specification

for

Citara: AI music generator

Version 1.0

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Revision History

Name	Date	Reasons for changes	Version
Nattanan Pimjaipong	4/2/2026	Fix some requirements & grammar & add use case & use case diagram	0.1
Nattanan Pimjaipong	10/2/2026	Specify stakeholders, add Open issues, Assumption and metric table. + Reformatting	0.2
Nattanan Pimjaipong	12/2/2026	Fix the stakeholders + formatting	0.3

Glossary

Terms	Definitions
AI	Artificial Intelligence
API	Application Programming Interface
Chromium	The open-source browser project that powers browsers like Google Chrome and Microsoft Edge
Google Auth	A third-party authentication service allowing users to log in using their Google account credentials
Metadata	Information provided by the system regarding a song's status, title, and the data used for its generation
Session	The time during which a User is actively using the system
SRS	System Requirements Specification
User	The person who operates the software product

1. Introduction

1.1 Purpose

Defining and describing the functions and specifications of the Cithara: AI music generator is the primary goal of this Software Requirements Specification (SRS).

This Software Requirements Specification illustrates, in clear terms, the system's primary uses and required functionality as specified by our customer.

1.2 Document Conventions

This document follows the standard typographical conventions established by IEEE for software engineering documentation. The following conventions are used to ensure clarity and interpretability:

Term	Description
SHALL	Indicates a strictly required function or constraint.
SHOULD	Indicates a high-priority but non-mandatory feature.

Requirements are categorized as follows:

Requirement Number	Description
US-XX	User stories
FR-XX	Functional Requirements
NFR-XX	Non-Functional Requirements

1.3 Intended audience

This document is intended for the following :

- Developer: who will design and develop the software
- Client: to ensure that the software implementation would produce as needed

1.4 Stakeholders

1. **End user:** The primary user who use this system to generate and listen to the song
2. **Product Owner:** The person who control the feature, clarify scope
3. **Developer:** The Developer who will create this system

1.5 Project Scope

The software being produced is called Cithara: AI music generator. This system is designed to provide an easy path to creating songs without any musical knowledge.

In Scope:

The software will include the following key components:

- The music generator AI, along with the form to prompt it.
- The library that contains the generated songs.
- The song player allows users to play a song.

Out of Scope

The system will not include these features.

- Publish the song online
- Edit the song
- Training the AI
- Admin dashboard and admin monitoring
- Searching for the song in the library
- Grouping songs as a playlist

1.6 Reference

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.

2. Overall Description

2.1 Product Perspective

Cithara is designed as an integrated AI music generation platform that serves as a bridge between human creativity and artificial intelligence. The system operates as a web-based application, ensuring broad accessibility through standard modern web browsers like Chrome and Firefox.

2.2 Product Functions

1. **AI song generator:** Generate the song using AI based on user's need
2. **Song Library:** Song Storage with each song's detail
3. **Music Player:** Play the Song user generate
4. **User Registration**

2.3 User classes

The platform will serve the following user classes:

1. Song Creators:
 - The user who wishes to create songs
 - The user who wants to share the songs that they created on this platform
2. Song listener
 - The user who listens to the song includes the song's owner and the persons being shared with

2.4 User Stories

ID	Description
US-01	As a Song creator , I want to fill in the title of the song so that I can recognize my song later.
US-02	As a Song creator , I want to fill in the occasion of the song so that I can get the song that fits my needs.
US-03	As a Song creator , I want to fill in the song genre so that I get the song that fits the genre I want.
US-04	As a Song creator , I want to fill in the mood of the song so that I get the song that fits with the mood I want.

US-05	As a Song creator , I want to input my own custom lyrics so that I can generate a song that matches my specific songwriting vision.
US-06	As a Song creator , I want to choose the singing voice type so I get the voice I prefer in the song I generate.
US-07	As a Song creator , I want to see the loading screen so that I know the song is being created and the webpage is not freezing.
US-08	As a Song creator , I want to see the notification when the song is generated so that I can know when the song is ready for me.
US-09	As a Song creator and Listener, I want to be able to confirm my identity using Google authentication so that I can log in and use the website.
US-10	As a Song creator , I want to be able to see my song library so that I can see/ listen to the song I created.
US-11	As a Song creator , I want to be able to read all songs' titles, status, and metadata so that I can know the information about the song I created.
US-12	As a Song creator , I want to be able to play the song I create so that I can listen to what I was creating.
US-13	As a Song creator , I want to be able to download the music I created as a file so that I can save and share it later.
US-14	As a Song creator , I want to edit my existing song's details so that I can change them to what I prefer.
US-15	As a Song creator , I want to be able to share my music so that others can listen to the song I created.
US-16	As a Song creator , I want to delete my song so that I can get rid of the song I don't want.
US-17	As a Song listener , I want a prominent Play/Pause button so that I can immediately start or stop the audio without searching for the control.
US-18	As a Song listener , I want to use Next and Previous buttons so that I can quickly cycle through my history of generated tracks without going back to a list view.

US-19	As a Song listener , I want a volume slide so that I can adjust the volume as I need.
US-20	As a Song listener , I want the player to auto-play the next track in my history when the current one ends so that I can listen to all my creations hands-free.
US-21	As a Song listener , I want to see the loading screen when the audio player is buffering so that I know that the app is working properly.
US-22	As a Song creator , I want to know if the song itself is unavailable so I do not confuse it with outside issues (speaker, network).

2.5 Operating Environment

1. Technical Environment

- A web-based application accessible via standard browsers (Chrome, Firefox)
- Fully responsive web design for access via smartphones and tablets
- Hosting options to be determined (on-premises, cloud, or hybrid)

2.6 Design and Implementation Constraints

- **Browser Compatibility:** Must be fully functional on Chromium and WebKit-based browsers.

2.7 Assumptions and Dependencies

- **Third-Party API Reliability:** It is assumed that the external AI music generation engine (not specified in the scope) will maintain an uptime of at least 99%
- **Google Auth Availability:** The system depends on Google's Authentication service being operational for third-party logins.
- **Browser Media Support:** It is assumed that the users' Chromium or Firefox browsers are configured to support the specific audio file formats (e.g., MP3, WAV) generated by the AI without requiring additional plugins.

2.8 Open Issue and Assumptions

- **Undefined Third-Party AI API:** The music generator API is not defined. So there is no detail about the AI token or how much request the system can send. *It will be assumed to use AI from Suno AI.*
- **File Format for Downloads:** the specific file format (e.g., .mp3, .wav, or .flac) when downloading the song has not been defined. *The file format will be assumed to be mp3*
- **Public URL Expiration:** The system generates "unique public URLs" for sharing. It is not yet decided if these links expire after a certain period or remain active indefinitely. *It will be assumed that there will not be any expiration date.*
- **Library Sorting:** there is no specification of how the library shall be sorted (Date Added, Date Modified, Alphabetically). *It will be assumed to be sorted by Date Added, Descending.*

3. System Features

3.1 AI song generators

- **Parameter input:** The system requires users to provide specific metadata, including the song's title, occasion, genre, mood, and preferred voice type.
- **Song generation:** Using the gathered parameters, the system generates a unique song in one click.

ID	Requirement
FR-01	The system SHALL require the song's title from the user
FR-02	The system SHALL require the song's occasion from the user
FR-03	The system SHALL require the song's genre from the user
FR-04	The system SHALL require the voice type from the user
FR-05	The system SHALL require the song's mood from the user
FR-06	The system SHALL get the song's lyrics from the user
FR-07	The system SHALL generate the song based on the information taken from the user.

3.2 Song Library

- **Song storage:** The system stores up to 20 generated songs per user, displaying them in a list format
- **Song details:** Users can view the specific data used for generation, including the link and song description
- **Song editing and deletion:** The system allows users to update song titles and descriptions or delete tracks they no longer wish to keep.
- **Song sharing and downloading:** Users can download songs to their local storage or generate unique public URLs to share their creations with others.

ID	Requirement
FR-08	The system SHALL provide the list of songs that the user has generated
FR-09	The system SHALL provide the song details: status, title, description, link, and all the data used for generating the song

FR-10	The system SHALL allow the user to delete the song
FR-11	The system SHALL allow the user to edit the song details: title and description
FR-12	The system SHALL download the song to the user's local storage
FR-13	The system SHALL provide the link for the user to share a song without downloading
FR-14	The system SHALL hold at most 20 songs for each user
FR-15	If the loading of the song fails, the system SHALL notify the user

3.3 Music Player

- **Playback Control:** Includes standard "Play" and "Pause" buttons for immediate audio control
- **Navigation:** Users can skip to the next track or return to a previous one to cycle through their generation history.
- **Audio Adjustment:** A volume slider is provided, allowing users to adjust audio levels as needed.
- **Auto-Play:** The player can automatically transition to the next track in the user's history for a hands-free listening experience.

ID	Requirement
FR-16	The system SHALL play selected audio when the user presses the play button
FR-17	The system SHALL pause active audio when the user presses the pause button
FR-18	The system SHALL skip to the next track when the user presses the next song button
FR-19	The system SHALL go back to the previous track when the user presses the previous song button
FR-20	The system SHALL provide volume adjustment
FR-21	The system SHALL play the next track automatically when the current track ends

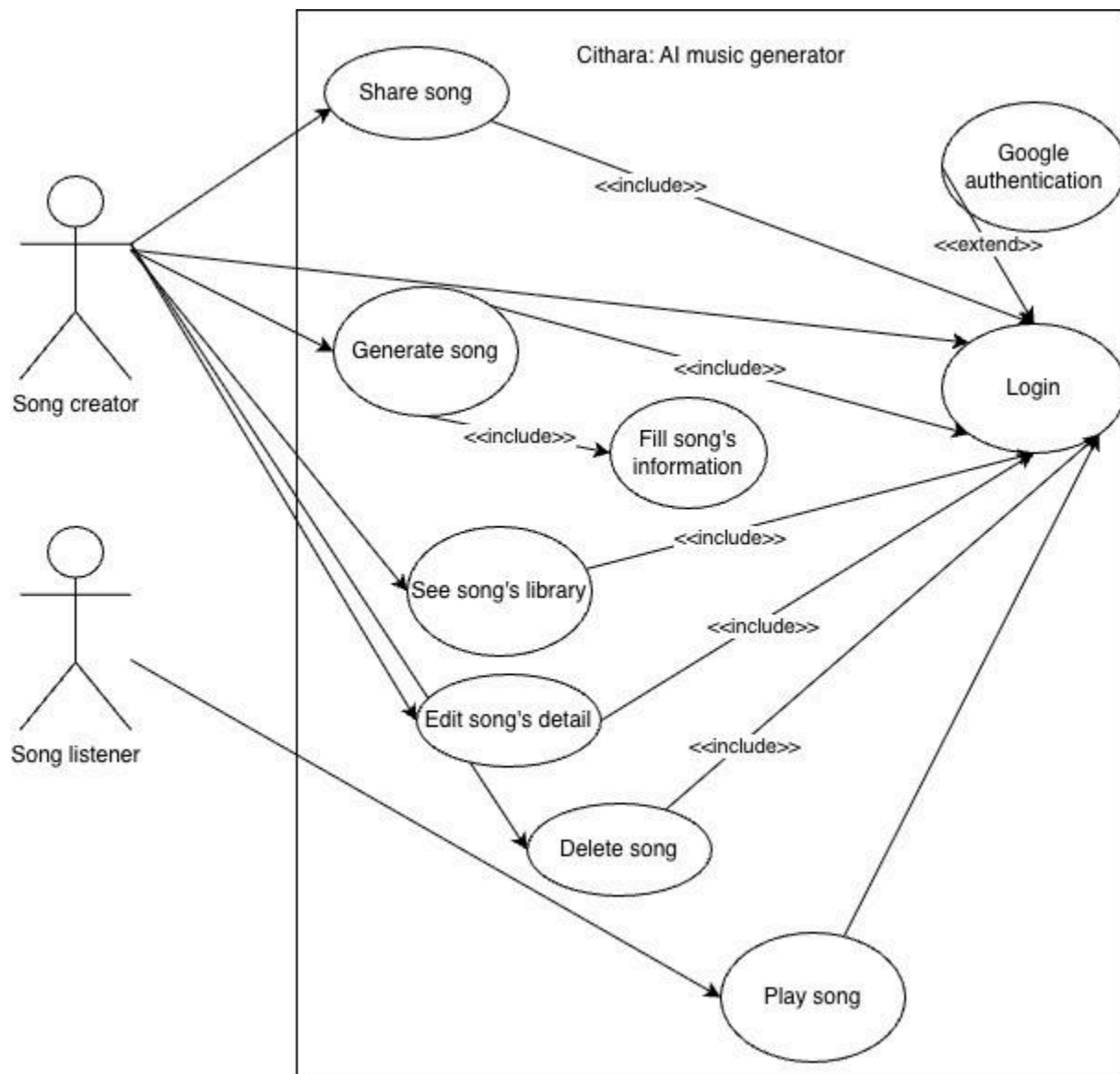
3.4 User Registration

- **Authentication:** Supports traditional local registration and login via username, email, and password, as well as third-party Google Authentication.
- **Account Recovery:** Provides a dedicated process for users to recover their credentials if lost.
- **Data Security:** Ensures all sensitive data is encrypted at rest and enforces strong password policies to protect user accounts.

ID	Requirement
FR-21	The system SHALL provide authentication via username and password
FR-22	The system SHALL provide local registration with name, username, email, and password
FR-23	The system SHALL support Google Auth integration.
FR-24	The system SHALL provide a credential recovery process.

4. Use case Modeling

4.1. Use case model



4.1.2 Use case Diagram Description

4.1.2.1 Usecase: Login

Field	Details
Use Case	Login
Actors	Song creator, Song listener
Type	Primary and essential
Description	Initiated when a user attempts an action that is restricted. The user is prompted to enter their username and password or use Google authentication to verify their identity.
Includes	None
Extends	Google authentication verified
Cross Ref	FR-21, FR-23, US-09
Use-Cases	None

4.1.2.2 Use Case: Generate song

Field	Details
Use Case	Generate song
Actors	Song creator
Type	Primary and essential
Description	The system processes the provided parameters to synthesize a unique musical track and adds it to the user's digital library.
Includes	Login, Fill song's information
Extends	None
Cross Ref	FR-07, US-07, US-08
Use-Cases	None

4.1.2.3 Use Case: Fill song's information

Field	Details
Use Case	Fill song's information
Actors	Song creator
Type	Primary and essential
Description	The user inputs specific metadata (title, occasion, genre, mood, lyrics, voice type) which acts as the prompt for the AI music engine.
Includes	None
Extends	None
Cross Ref	FR-01 through FR-06, US-01 through US-06
Use-Cases	None

4.1.2.4 Use Case: See song's library

Field	Details
Use Case	See song's library
Actors	Song creator
Type	Primary and essential
Description	The user views their personal repository of generated tracks, including status indicators and links to track details.
Includes	Login
Extends	None
Cross Ref	FR-08, FR-09, US-10, US-11
Use-Cases	Edit song's detail, Delete song, Share song

4.1.2.5 Use Case: Edit song's detail

Field	Details
Use Case	Edit song's detail
Actors	Song creator
Type	Primary and essential
Description	Allows the user to modify metadata (title/description) of an existing track to keep their library organized.
Includes	Login
Extends	None
Cross Ref	FR-11, US-14
Use-Cases	None

4.1.2.6 Use Case: Delete song

Field	Details
Use Case	Delete song
Actors	Song creator, Song listener
Type	Primary and essential
Description	The user removes a track from the system. This is often used to manage the 20-song account limit.
Includes	Login
Extends	None
Cross Ref	FR-10, US-16
Use-Cases	None

4.1.2.7 Use Case: Share song

Field	Details
Use Case	Share song
Actors	Song creator
Type	Primary and essential
Description	The system generates a public URL for a song and copies it to the user's clipboard for external distribution.
Includes	Login
Extends	None
Cross Ref	FR-13, US-15
Use-Cases	None

4.1.2.8 Use Case: Play song

Field	Details
Use Case	Play song
Actors	Song creator, Song listener
Type	Primary and essential
Description	The user interacts with the music player to listen to audio, adjust volume, and navigate through the playlist.
Includes	Login
Extends	None
Cross Ref	FR-16 through FR-20, US-12, US-17 through US-20
Use-Cases	None

4.2. Dressed up use case — Song generation

Primary Actor: Song Creator

Stakeholders and Interest:

- The Song creator wants to create a song that aligns with their need in one click

Precondition:

- The Song creator is verified and authenticated by the system.

PostCondition:

- The system successfully generate song and put in into the user's library

Main Success scenario:

1. The User fills in all the information needed [Alt1:the user doesn't fill out all requirement field]
2. The User pressed the song generate button [Alt2: user puts the unknown language in]
3. The system scans the content the user puts in and passes the scan [Alt3: The system detects offensive content]
4. The system generate the song while notify the user [Alt4: fail to generate][Alt5: the library is full]
5. The system put the show that the song is done and ready in their library
6. The user checks out the library with the generation song inside

Alternatives flow

ALT1. The user doesn't fill out all requirement field

1. The user presses the generate button
2. The system highlights the box that hasn't been filled (first one)
3. The user repeats until all the requirement fields are filled
4. Go back to the Main Success scenario 2

ALT2. The system detects offensive content

1. The system found the offensive words (sexism, racism)
2. The user is notified that the word in that category is not allowed
3. Go back to the Main Success scenario 1

ALT3. The system failed to generate

1. The system will try to send the request again
2. The system success and goes back to the Main Success scenario 5

ALT4. The system cannot load the library

1. The system will notify the user and test again

ALT5. The library is full

1. The system will not let the user generate the song, and shall notify the user.
2. The user checks out the library
3. The system shows the list of songs
4. The user chooses one song and deletes (3.2.4 brief use case — download song)
5. Go back to the Main Success scenario 4

ALT6. The system failed to generate again

1. The system will no longer send the request and notify the user
2. Go back to the main success scenario 2 with the information filled

Exceptions:

- If there is any progress that took time, show the loading page. If the process runs in the background, notice the user on the side of the screen.
- If an error happens(after already trying once), keep the user input, then notify the user.

Special Requirements:

1. The generation timeout: 10 min
2. If the song generation took more than 1 min, download in the background
3. The song will generate up to 3 songs at the same time
4. The generation will happen only when the user opens the page; if not, pause the generation

4.3. Brief use case — View song's details

- The user navigates to the Library section from the main page.
- The system shows the list of the user-generated songs.
- The user clicks the "3 dots" icon on the specific song they wish to see its details.
- The system show the options(share, view, edit, download)
- The user presses View
- The system shows the song details (song title, occasion, genre, voice type, mood, lyrics, metadata, date created, date edited)

4.4. Casual use case — Download song

Main-Flow

- The user opens the library
- The system shows the list of the user-generated songs.
- The user clicks the "3 dots" icon on the specific song they wish to download.
- The system shows options (share, view, edit, download)
- The user choose download
- The system packages the audio file and saves it directly to the user's local storage.

Alternative Flows / Exceptions

- **A1: Empty Library:** If the user has no songs, the system displays an "Empty Library" message with a prompt to generate a new song.
- **A2: Storage Permission:** If the browser or device denies storage access, the system prompts the user to "Allow Permissions" to complete the download.
- **A3: Connection Interrupted:** If the network fails during the download, the system shows a "Download Failed" notification with a Retry button.

4.5. Casual use case — Share song

Main Flow

- The user navigates to the Library section from the main page.
- The system shows the list of the user-generated songs.
- The user clicks the "3 dots" icon on the specific song they wish to share.
- The system displays the options menu (Share, View, Edit, Detail, Download).
- The user chooses the Share option.
- The system generates a unique public URL for the song and automatically copies it to the user's clipboard.
- The system displays a "Link Copied" notification to the user.

Alternative Flows / Exceptions

- **A1: Share Menu Failure:** If the link cannot be generated due to a network error, the system displays a "Sharing Unavailable" message and suggests trying again later.

4.6. Casual use case — Delete song

Main Flow

1. The user navigates to the Library section from the main page.
2. The system displays the list of the user's generated songs.
3. The user clicks the "3 dots" icon on the song they want to remove.
4. The system displays the options menu.
5. The user chooses the Delete option.
6. The system displays a confirmation pop-up asking, "Are you sure you want to delete this song?"
7. The user selects "Yes/Confirm."
8. The system deletes the song from the database and removes it from the user's view.

Alternative Flows / Exceptions

A1:Cancel Deletion: If the user selects "Cancel" at the confirmation prompt, the system closes the pop-up, and no changes are made.

A2: Deletion Error: If the server fails to delete the song (e.g., database timeout), the system shows an error message: "Unable to delete song at this time. Please try again."

5. Other Nonfunctional Requirements

This section specifies the non-functional requirements for the Cithara: AI music generator. These requirements define the quality attributes and constraints that the system must satisfy to meet stakeholder expectations.

5.1. Performance Requirements

ID	Requirement
NFR-01	The system SHALL provide page load times of less than 2 seconds for standard operations under normal load conditions.
NFR-02	The system SHALL handle at least 1000 requests per minute
NFR-03	The system SHALL support Chromium and Firefox browsers
NFR-04	The system SHALL be able to handle 50 GB of data
NFR-05	If the song generation takes longer than 1 minute, the system SHALL generate the song in the background.
NFR-06	The system SHALL not take time to generate the song longer than 10 minutes per request. If so, mark as generation fails.
NFR-07	The system SHALL play music after the play button is pressed in 0.2 seconds.

5.2 Usability and Accessibility Requirements

ID	Requirement
NFR-08	The system SHALL implement responsive design to support various screen sizes and devices.
NFR-09	The system SHALL provide a consistent and intuitive user interface across all functions.
NFR-10	The system SHALL ensure sufficient color contrast for text and interactive elements.

NFR-11	The system SHALL provide support only for the English language
NFR-12	The system SHALL generate at most 3 songs at a time
NFR-13	The system SHALL restrict the user from using offensive words in the lyrics (sexism, racism)
NFR-14	The music player SHALL be able to navigate through the keyboard
NFR-15	The system SHALL show the loading screen/icon every time the system takes time to generate or load the song
NFR-16	The system SHALL keep the user input when the system fails to generate the songs.

5.3 Security Requirements

ID	Requirement
NFR-16	The system SHALL enforce strong password policies, including minimum length, complexity, and regular password changes.
NFR-17	The system SHALL encrypt all sensitive data at rest using industry-standard encryption algorithms
NFR-18	The system SHALL support OAuth 2.0 and OpenID Connect for third-party authentication where applicable.
NFR-19	The system SHALL implement secure session management with 30-minute timeout settings.

5.4. Reliability and Availability

ID	Requirement
NFR-20	When the song generation failed, the system SHALL try to send the request once again.
NFR-21	When an error occurs, the system SHALL notify the user with a meaningful error message.

Appendix A: Traceability metrics

User Story ID	User Story Description	Functional Requirement (FR) / Specification
US-01	Fill in the song title	FR-01: System SHALL require song title
US-02	Fill in the occasion	FR-02: System SHALL require song occasion
US-03	Fill in the song genre	FR-03: System SHALL require song genre
US-04	Fill in the song mood	FR-05: System SHALL require song mood
US-05	Input custom lyrics	FR-06: System SHALL get song lyrics from user
US-06	Choose singing voice type	FR-04: System SHALL require voice type
US-07	See loading screen during creation	NFR-15: SHALL show loading screen/icon
US-08	Notification when song is ready	Use Case 3.2: System notifies user during generation
US-09	Google authentication login	FR-23: SHALL support Google Auth integration
US-10	See song library	FR-08: See song library
US-11	Read titles, status, and metadata	FR-09: Read song's title, status, and metadata Use Case 3.3: View song's detail
US-12	Play created songs	FR-16: Music player functionality

US-13	Download music as a file	FR-12: SHALL download song to local storage Use Case 3.4: Download song
US-14	Edit existing song details	FR-11: SHALL allow editing title and description
US-15	Share music via link	FR-13: SHALL provide link to share without downloading Use Case: 3.5 Share song
US-16	Delete unwanted songs	FR-10: SHALL allow user to delete the song Use Case 3.6: Delete Song
US-17	Prominent Play/Pause button	FR-17: Play/Pause button requirement
US-18	Next and Previous buttons	FR-18: Navigation buttons requirement
US-19	Volume slider	FR-19: Volume slider requirement
US-20	Auto-play next track	FR-21: Auto-play next track requirement
US-21	Loading screen when buffering	NFR-15: SHALL show loading screen when loading song
US-22	Notification if song is unavailable	FR-15: If loading fails, system SHALL notify user